

FASTLANE ***ENGINE***

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Quelques exemples



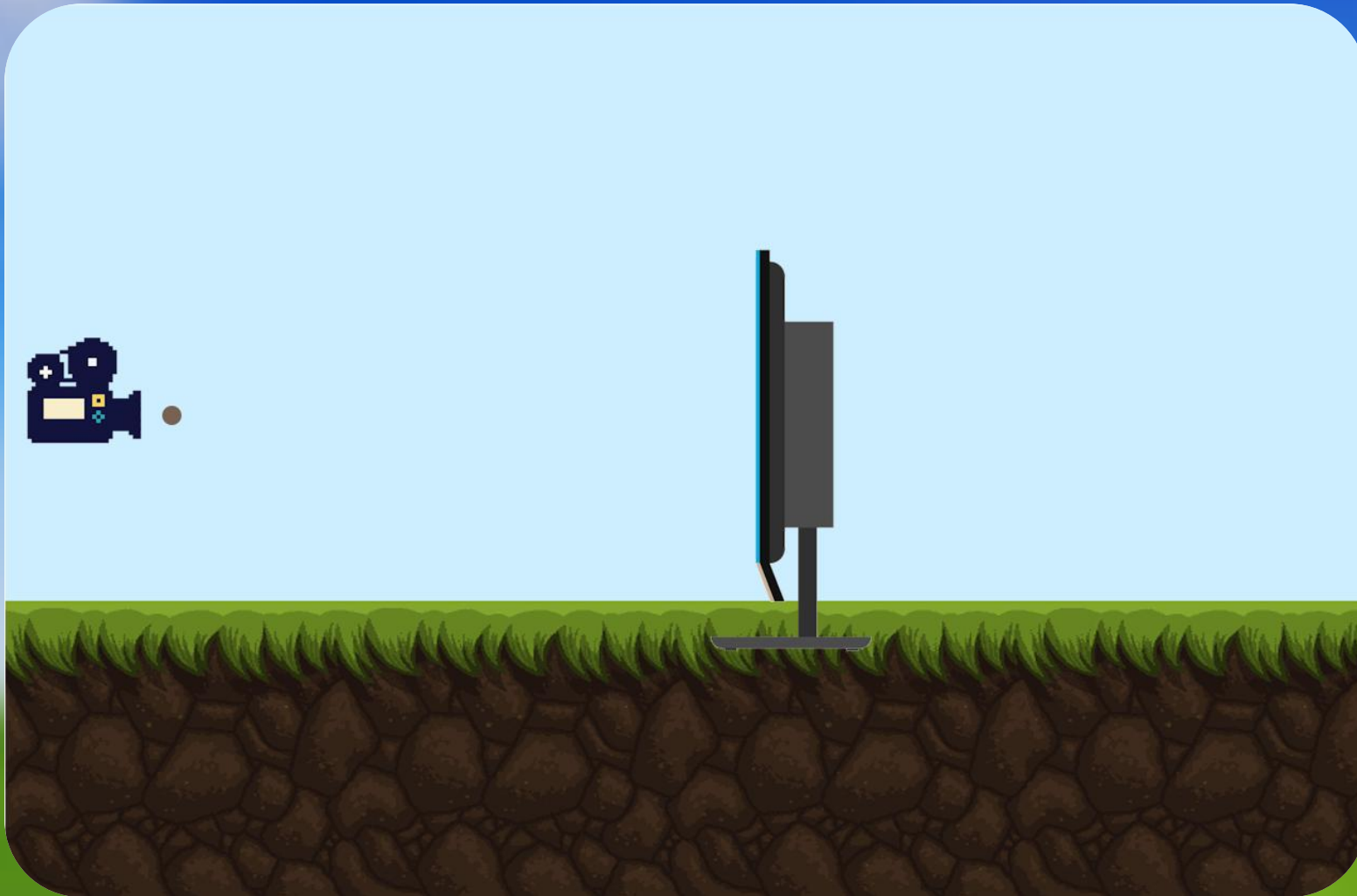
OutRun (1986)



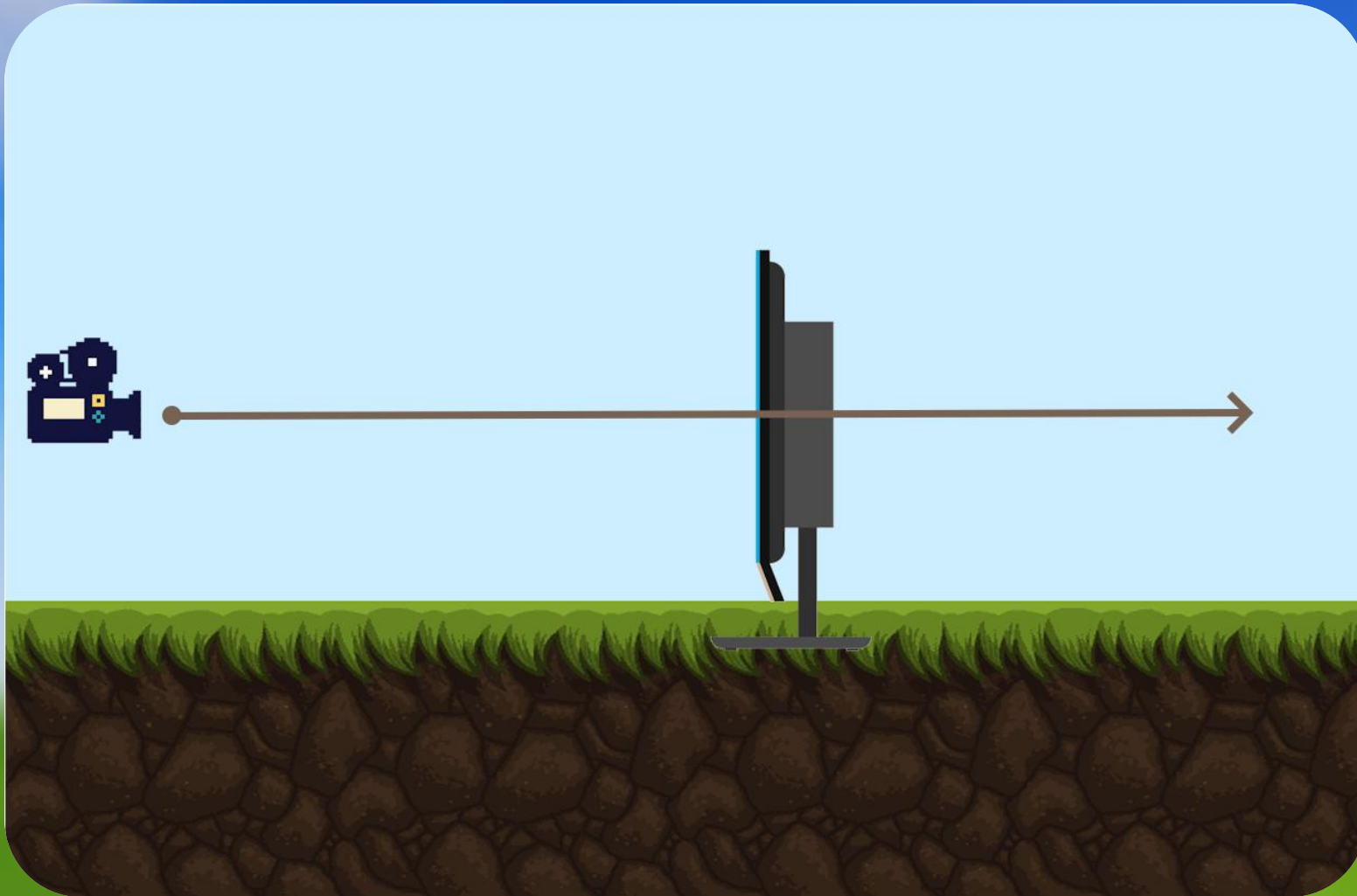
Super Mario Kart (1992)

Concept de base

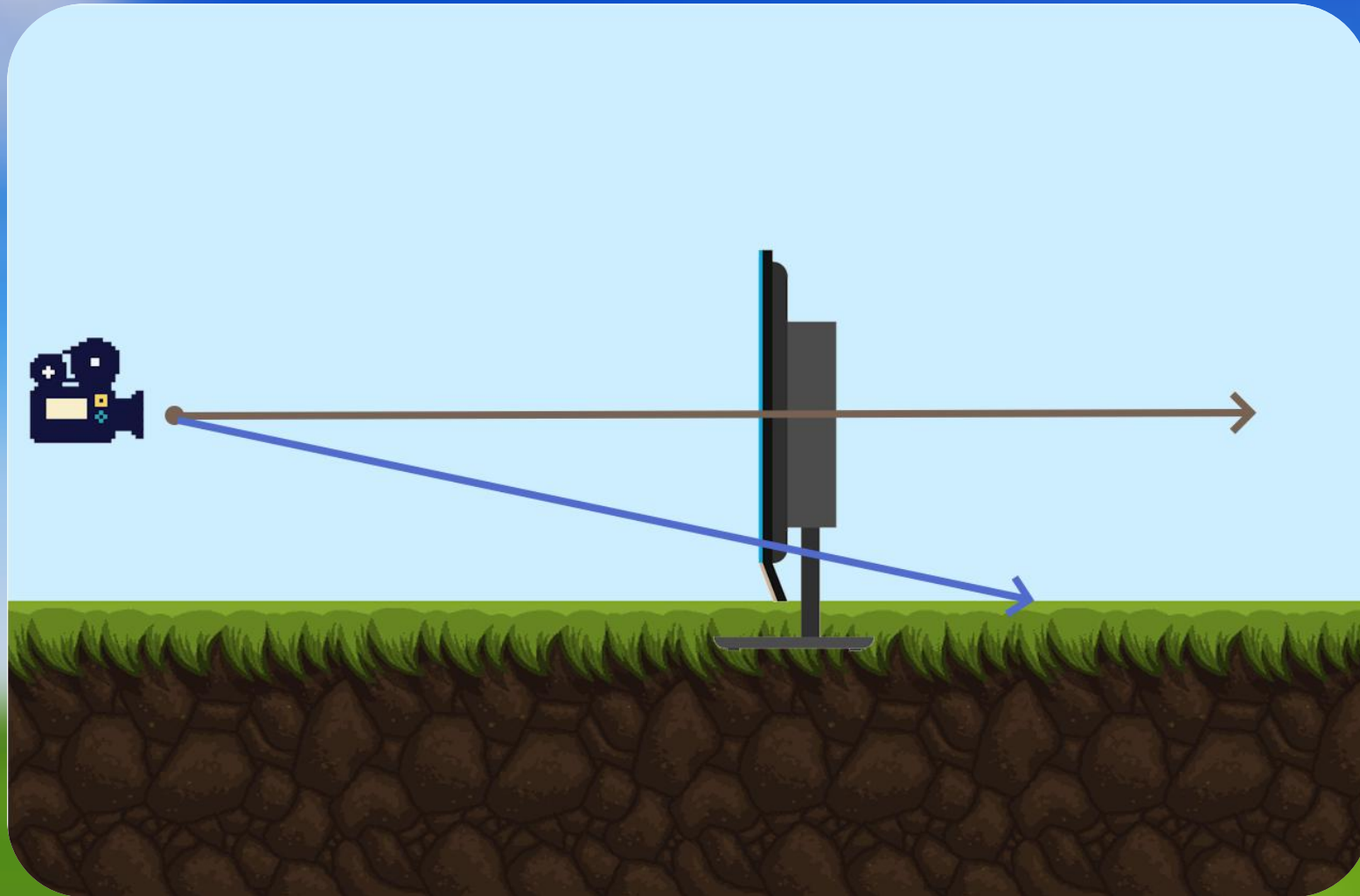
Concept de base



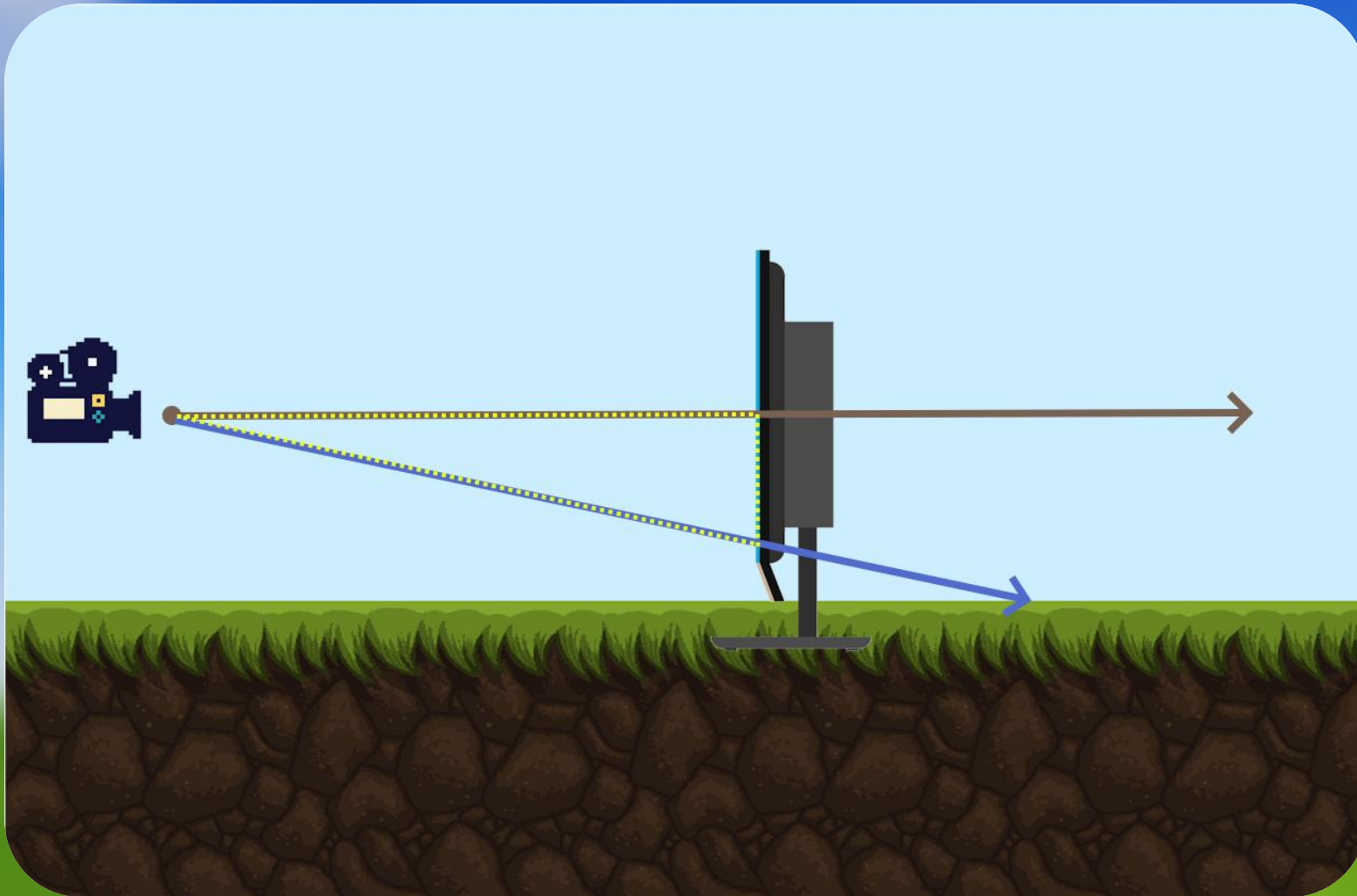
Concept de base



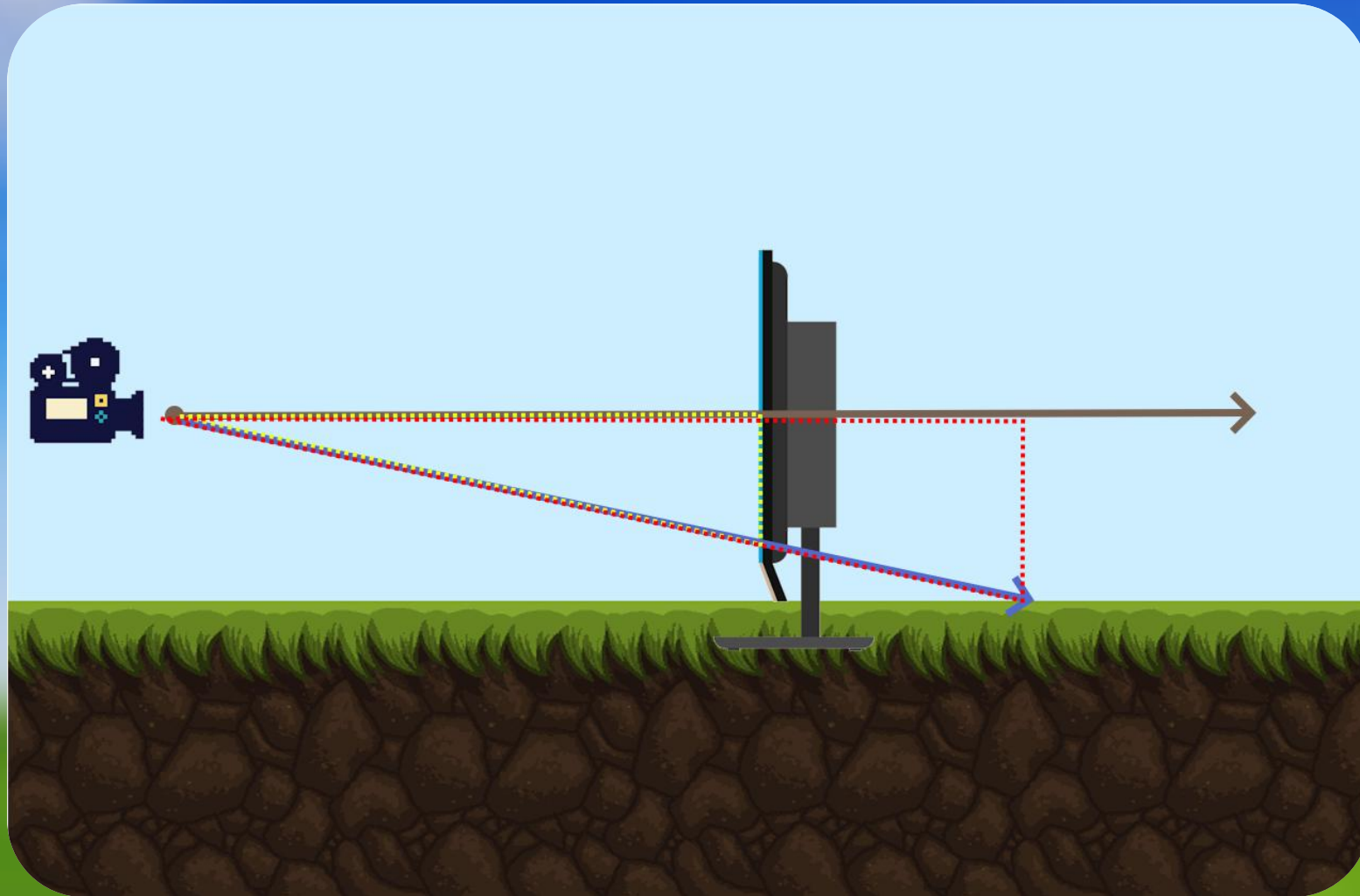
Concept de base



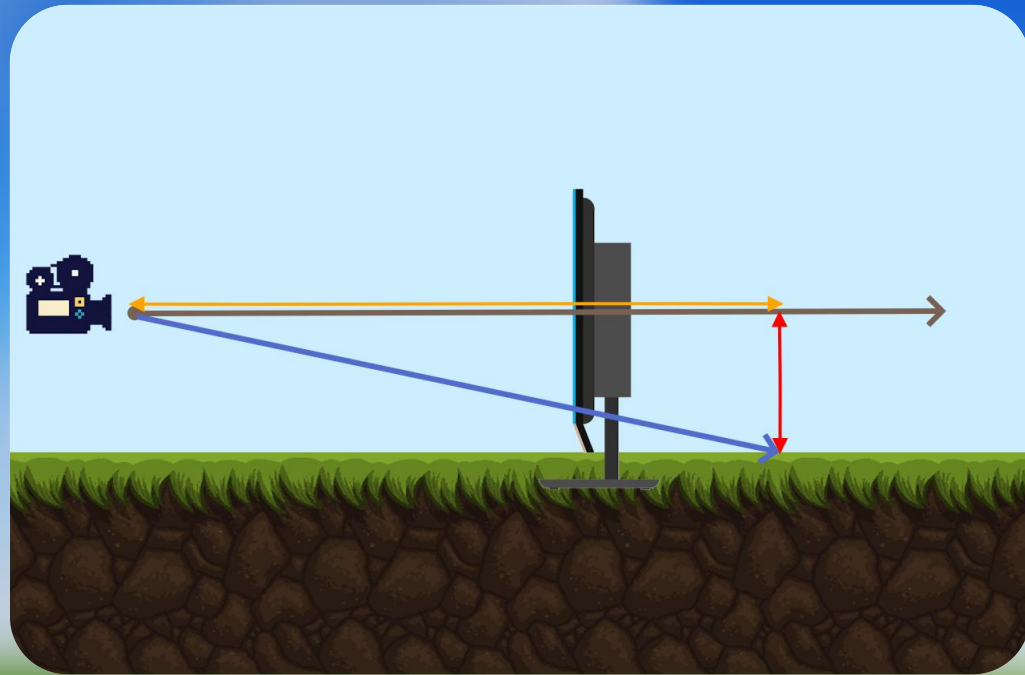
Concept de base



Concept de base



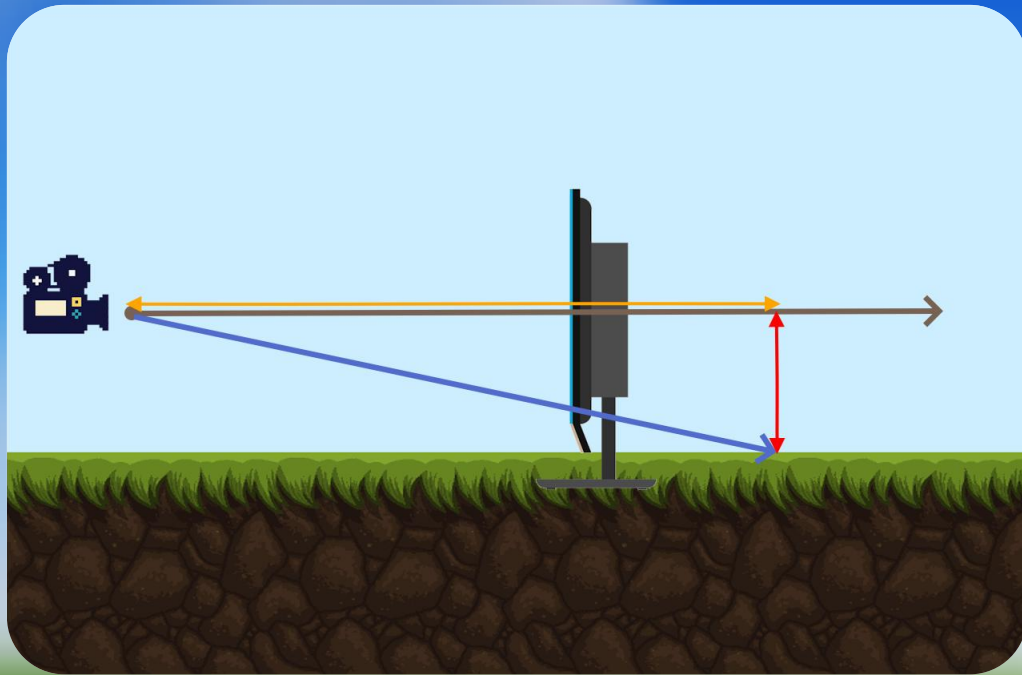
Concept de base



$$\frac{height}{2}$$

z

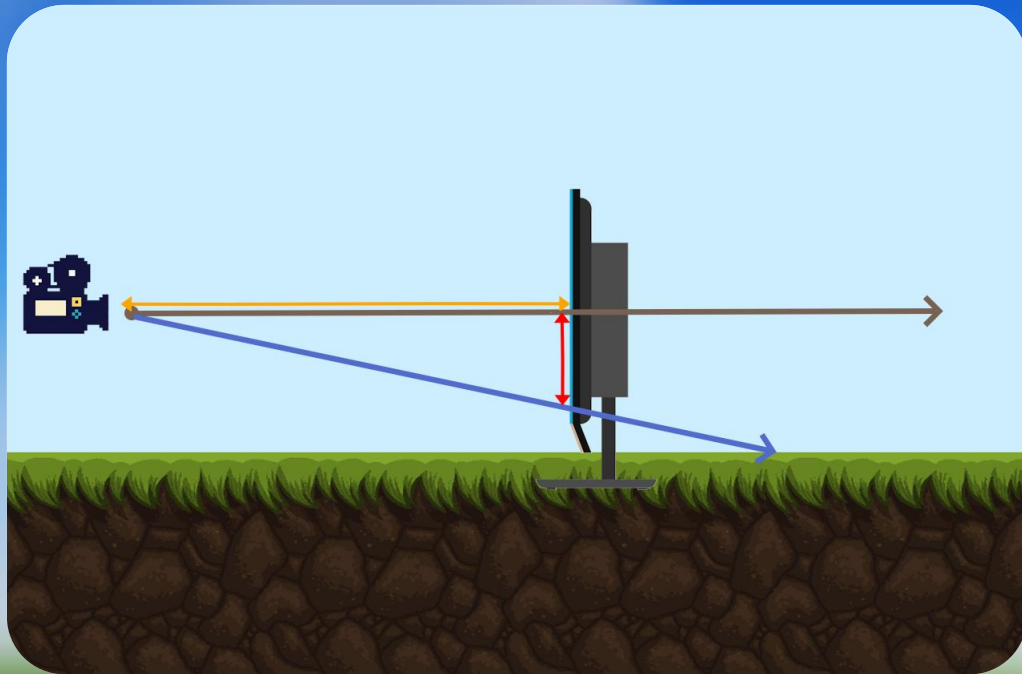
Concept de base



cam_height

Z

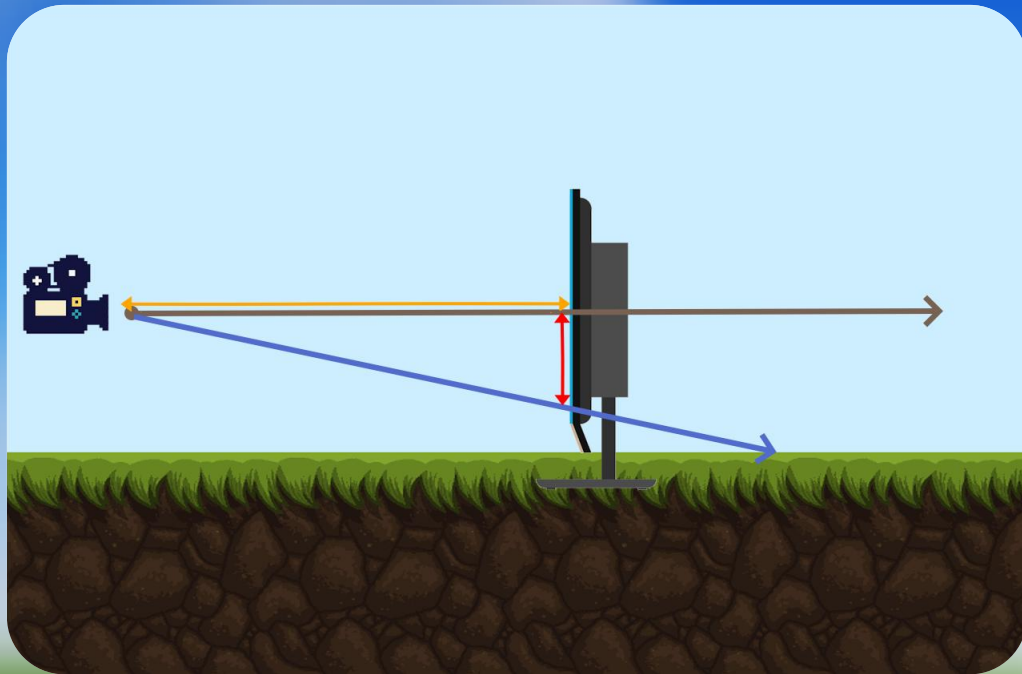
Concept de base



$$\frac{cam_height}{z}$$

$$\frac{y_screen - height / 2}{distance}$$

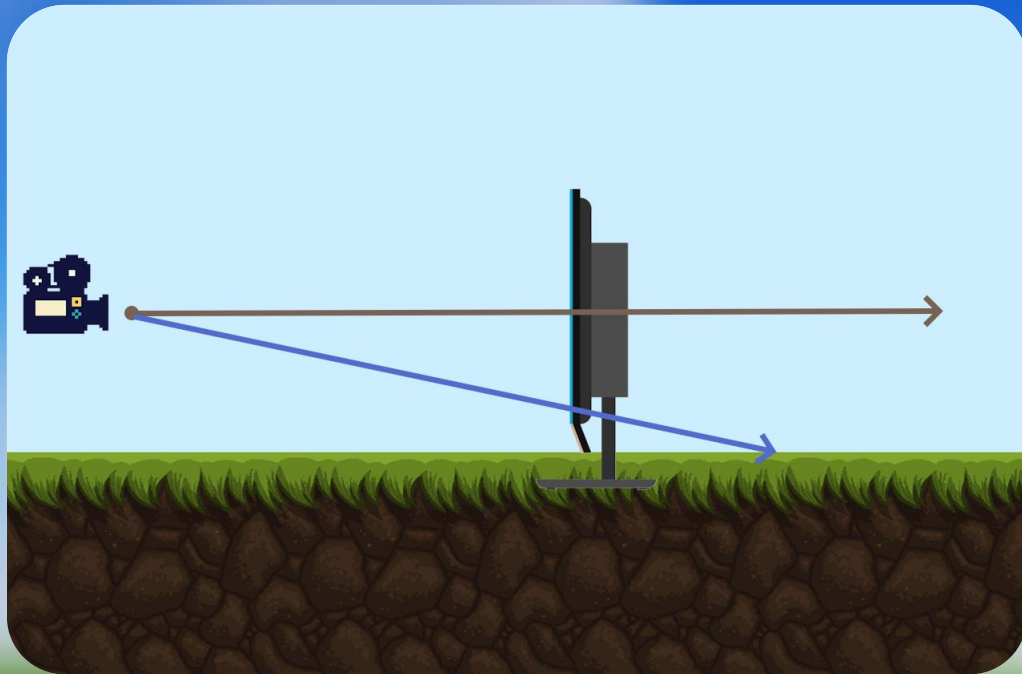
Concept de base



$$\frac{cam_height}{z}$$

$$\frac{y_screen - horizon}{distance}$$

Concept de base



$$\frac{cam_height}{z}$$

$$\frac{y_screen - horizon}{distance}$$

Concept de base

Scanline

$$z = \frac{cam_height \cdot distance}{y_screen - horizon}$$

Concept de base

Projection

$$y_{screen} = \frac{cam_height \cdot distance}{z} + horizon$$

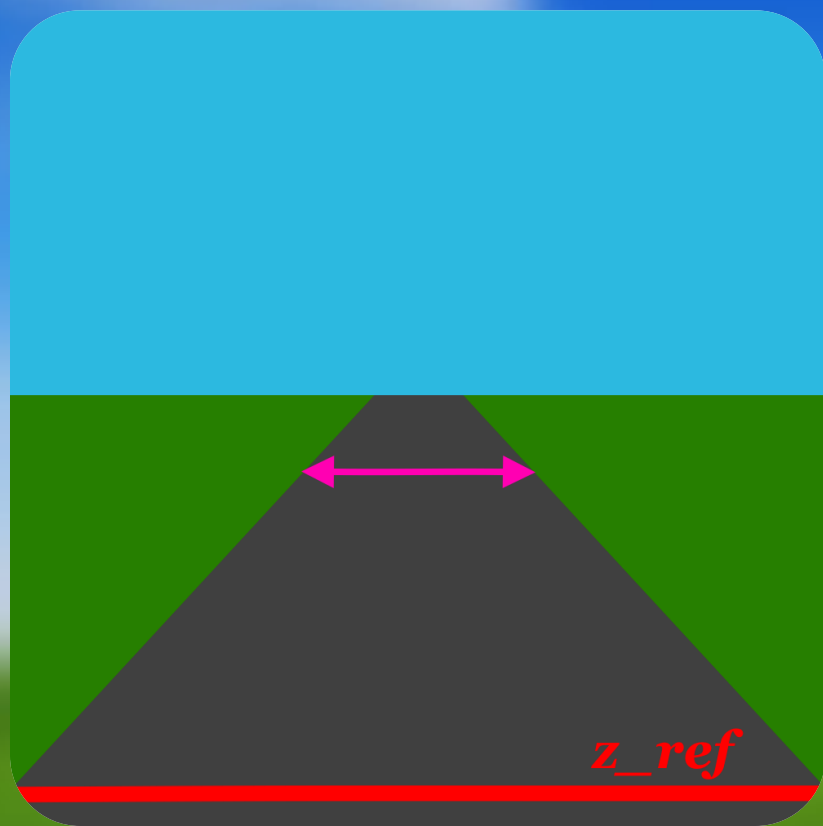
Concept de base

Projection



Concept de base

Projection



$$road_width_proj = \frac{z_{ref} \cdot road_width}{z}$$

Virages

Création des routes

📄 Données de piste

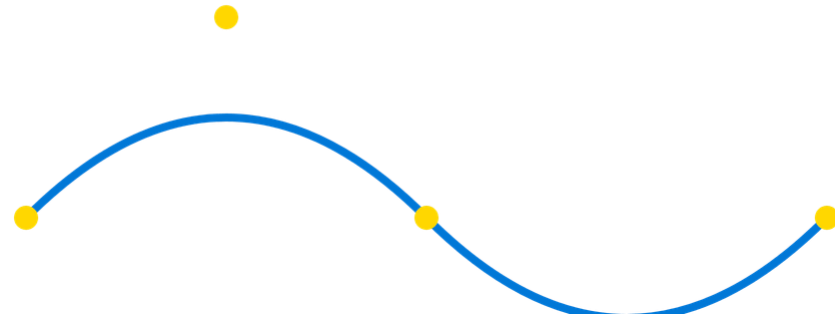
Lecture d'un fichier **txt** au lancement :

- **Altitude** : Variation verticale.
- **Déport** : Position horizontale du centre.

Calcul continu via `trackXAt(z)` avec interpolation précise entre les points de données.

```
roads > track.txt
1  0 0.000000
2  4 0.000000
3 18 0.000000
4 40 0.000000
5 71 0.000000
```

Lissage Catmull-Rom



Évite les "angles" brusques pour une trajectoire fluide et naturelle.

Virages

Gestion de la caméra



POSITIONNEMENT

Gestion de la position latérale du joueur.

Simule le décalage de la voiture sur la largeur de la route pour une conduite active.



SUIVI (LATFOL)

Compensation automatique basée sur **trackXAt(roadPos)**.

Maintient la route centrée à l'écran malgré les virages, assurant un confort visuel optimal.



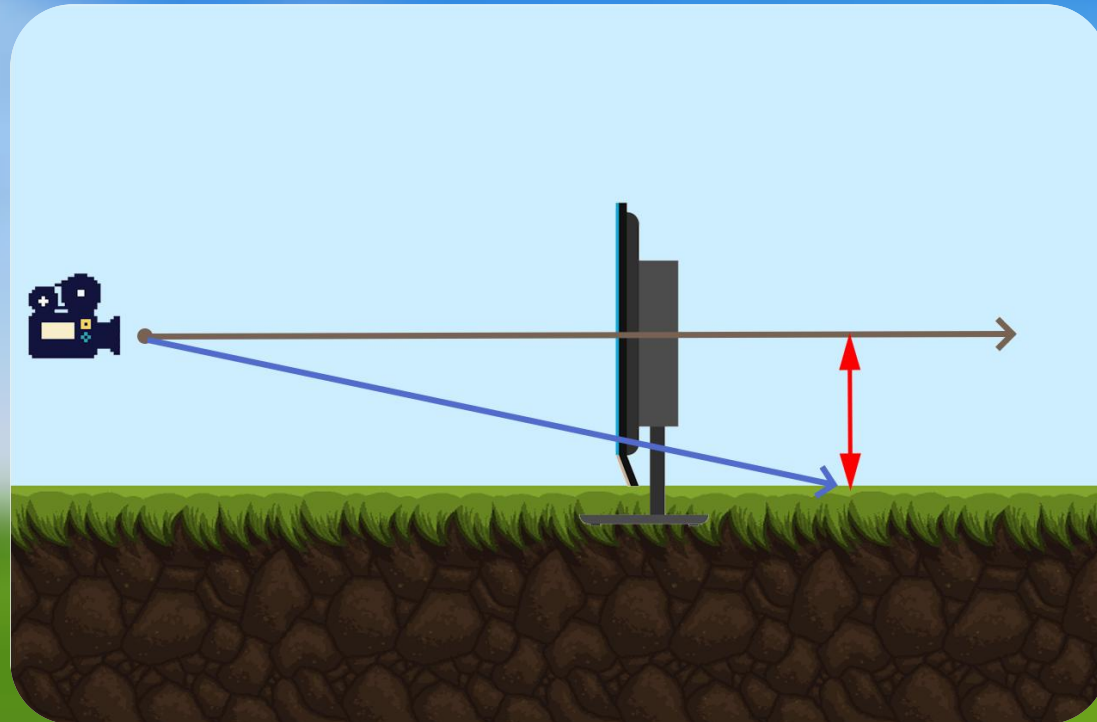
ANTICIPATION

Mode **yawed** utilisant un point distant .

Incline la projection pour "regarder" dans le virage avant d'y entrer, renforçant le réalisme.

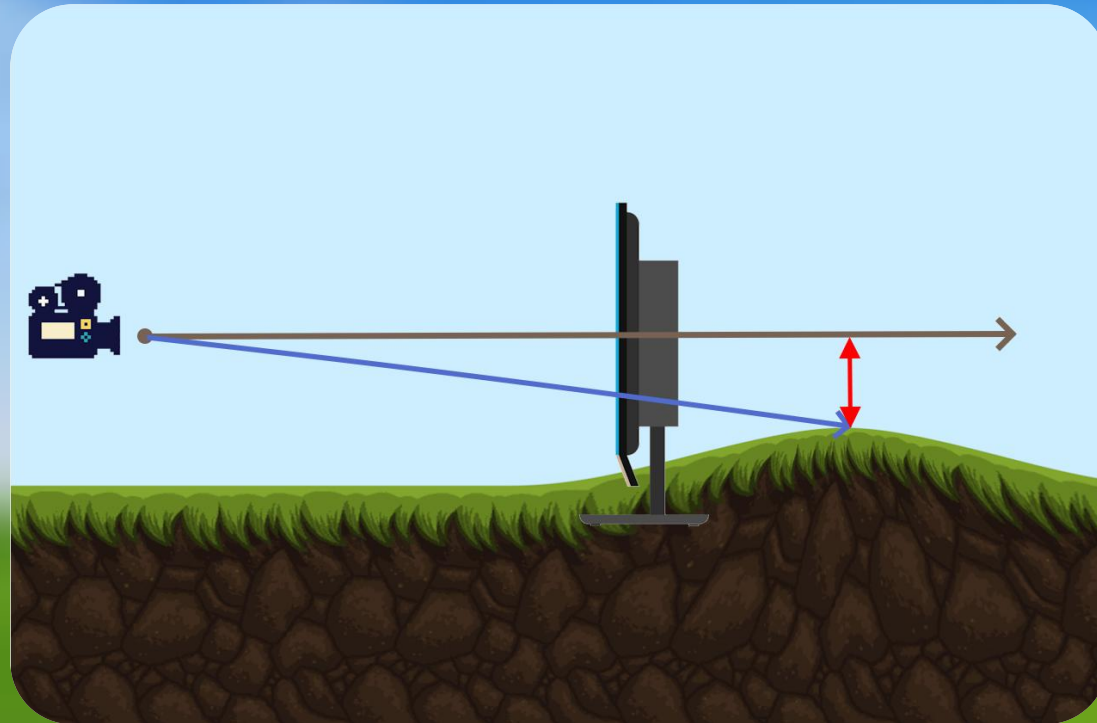
Altitude

$$y_{\text{screen}} = \frac{\textit{cam_height} \cdot \textit{distance}}{z} + \textit{horizon}$$



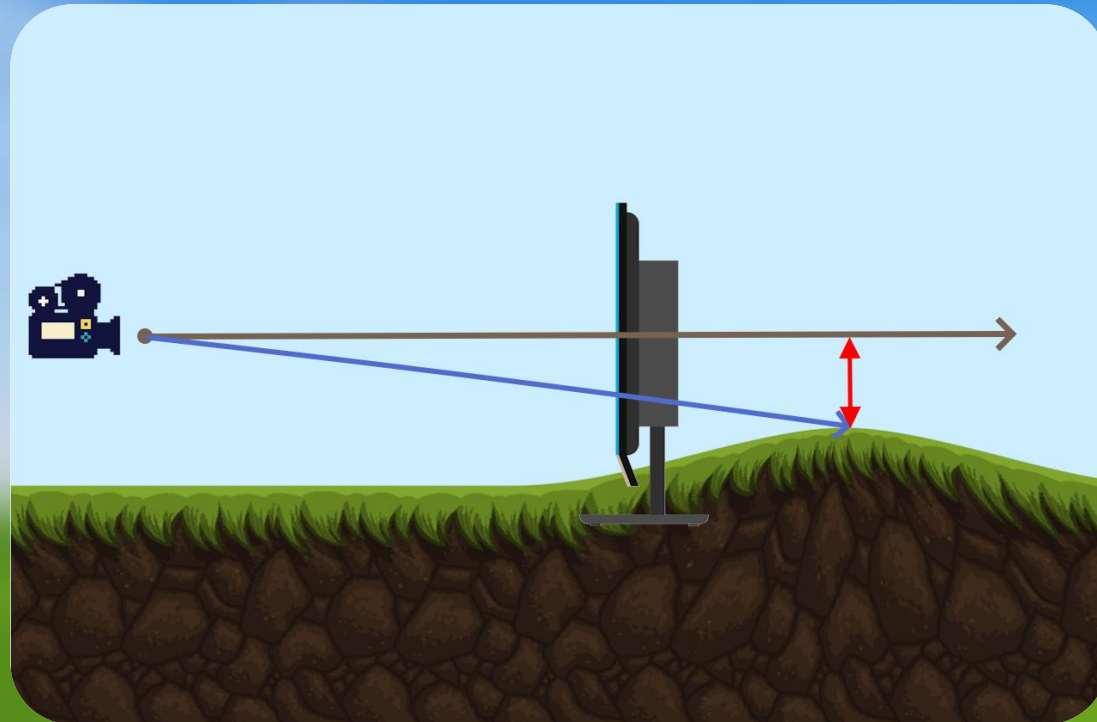
Altitude

$$y_{\text{screen}} = \frac{\text{cam_height} \cdot \text{distance}}{z} + \text{horizon}$$

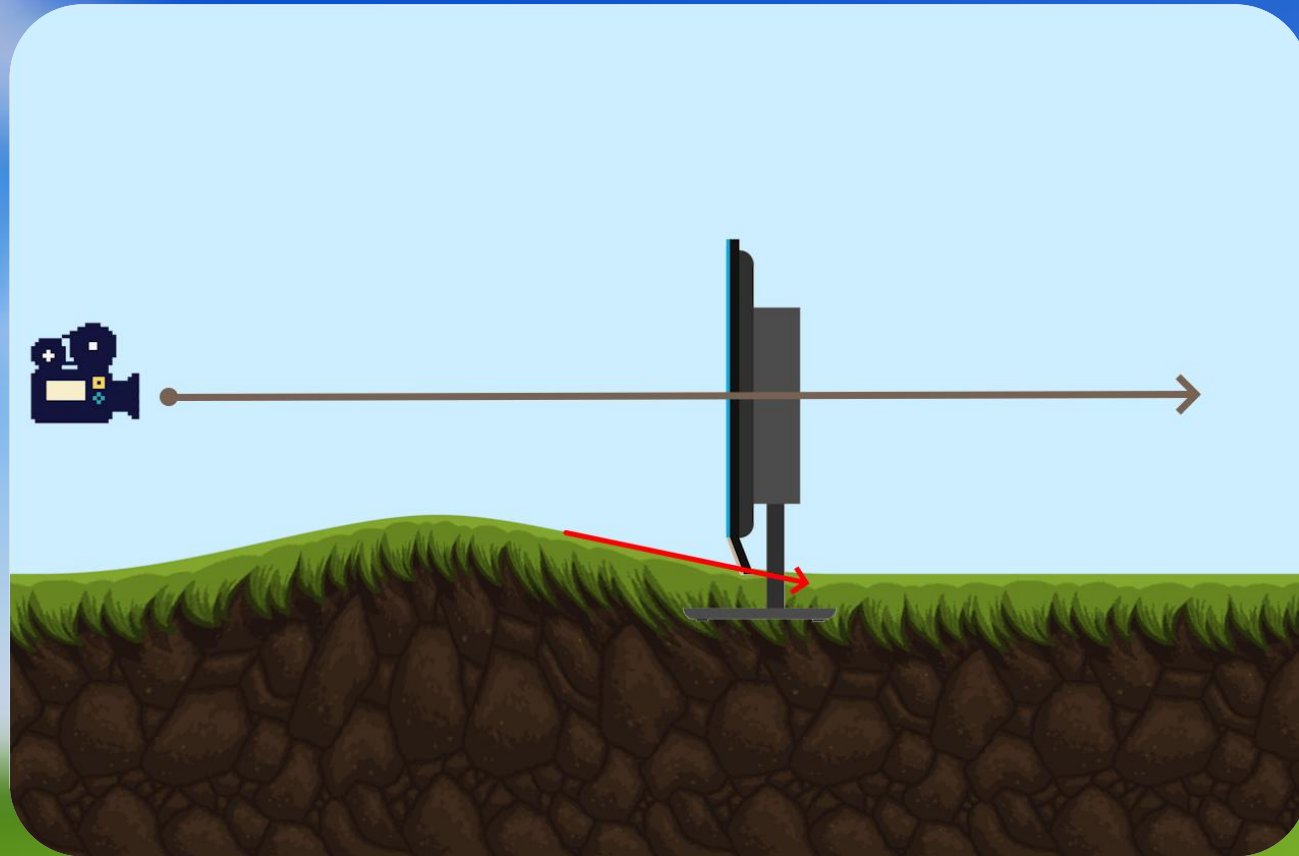


Altitude

$$y_{\text{screen}} = \frac{(cam_height - altitude(z)) \cdot distance}{z} + horizon$$



Altitude



Altitude

